Machine Learning Model Comparison Report

Corrosion Rate Prediction · Regression Model Evaluation

A comprehensive visual comparison of four regression models covering prediction performance, residual diagnostics, and model evaluation metrics (R^2 and RMSE).

■ Model Performance Summary

Relative performance of all four models based on visual inspection of prediction plots and error charts.

Linear Regression	~0.97	~0.28	Tight (±0.8)	■ 1st
XGBoost	~0.96	~0.36	Moderate (±1.0)	■ 2nd
Gradient Boosting	~0.96	~0.38	Moderate (±1.0)	■ 3rd
Random Forest	~0.93	~0.53	Wide (±1.5)	4th

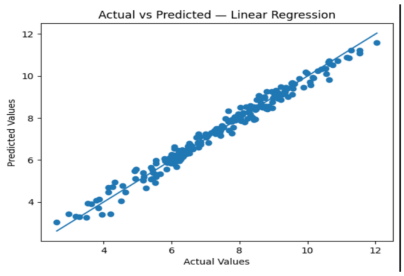
1 · Actual vs Predicted Plots

Scatter plots comparing actual versus predicted corrosion rate values. Points close to the diagonal reference line indicate high accuracy. Linear Regression and Gradient Boosting show the tightest clustering.

1.1 Linear Regression · XGBoost

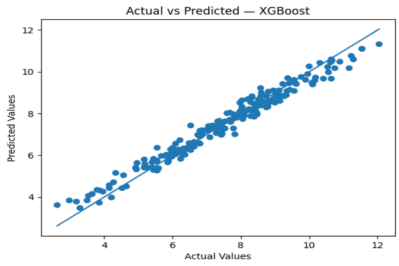
prediction. It includes prediction performance, residual analysis, and overall model evaluation metrics.

Actual vs Predicted — Linear Regression



Actual vs Predicted — XGBoost

Fig 1a — Linear Regression: Actual vs Predicted

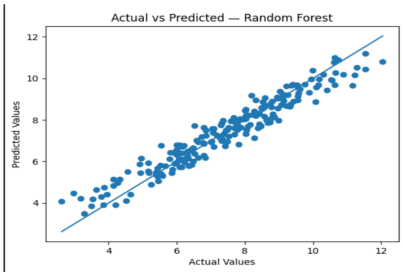


Actual vs Predicted — Gradient Boosting

Actual vs Predicted — Gradient Boosting

Fig 1b — XGBoost: Actual vs Predicted

1.2 Gradient Boosting · Random Forest



Residual Plot — Linear Regression

Fig 1c — Gradient Boosting: Actual vs Predicted

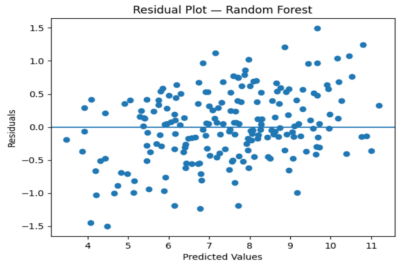
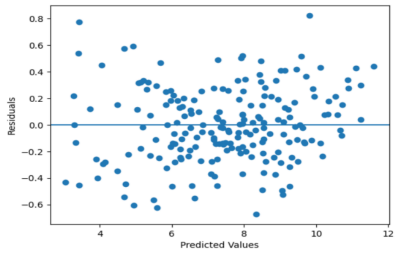


Fig 1d — Random Forest: Actual vs Predicted

2 · Residual Analysis

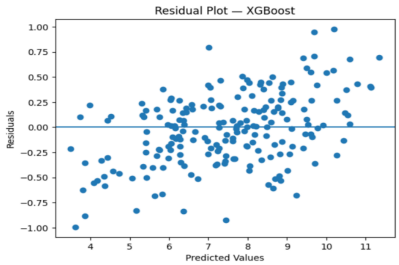
Residual plots reveal systematic errors and heteroscedasticity. Ideally residuals scatter randomly around zero. Linear Regression has the narrowest band (~±0.8); Random Forest the widest (~±1.5).

2.1 Linear Regression · XGBoost



Residual Plot — XGBoost

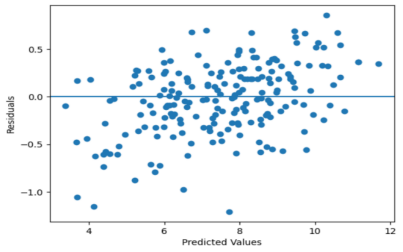
Fig 2a — Linear Regression: Residuals



Residual Plot — Gradient Boosting

Fig 2b — XGBoost: Residuals

2.2 Gradient Boosting · Random Forest



Residual Plot — Random Forest

Fig 2c — Gradient Boosting: Residuals

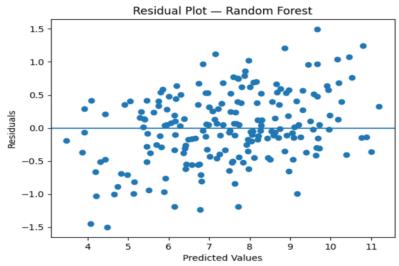


Fig 2d — Random Forest: Residuals

3 · Model Accuracy & Error Comparison

Bar charts aggregating model performance across the full test set. All models achieve $R^2 > 0.93$. Linear Regression achieves the lowest RMSE (~0.28), making it the most precise model despite its simplicity.

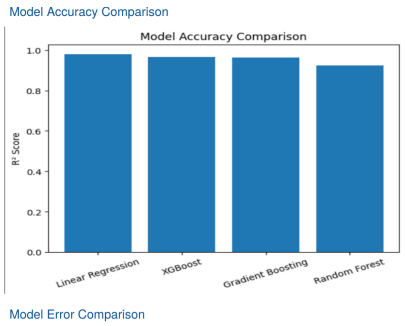


Fig 3a — R² Score Comparison (higher = better)

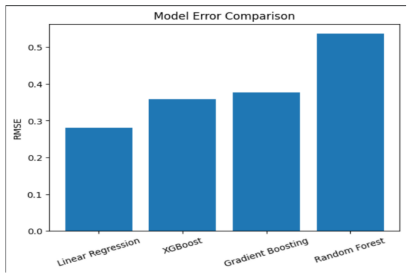


Fig 3b — RMSE Comparison (lower = better)

4 · Conclusions & Recommendations

Finding	Detail
Best overall model	Linear Regression — lowest RMSE (~0.28) and tightest residuals
Best ensemble model	XGBoost — strong balance of accuracy and error control
Weakest performer	Random Forest — highest RMSE (~0.53) and widest residual spread
Next steps	Tune XGBoost/GB hyperparameters; collect more extreme-value data; run feature importance analys